UCT511: COVERSATIONAL SYSTEMS

L	T	P	Cr
2	1	2	3.5

Course Objectives: The objective of the course is to enable attendees to acquire knowledge on chatbots and its terminologies. Work with ML Concepts and different algorithms to build custom ML Model. Better understand on Conversational experiences and provide better customer experiences.

Fundamentals of Conversational Systems:
Introduction: Overview, Case studies, Explanation about different modes of engagement for a human being, History and impact of AI Underlying technologies: Natural Language Processing, Artificial Intelligence and Machine Learning, NLG, Speech-To-Text, Text-To-Speech, Computer Vision etc. Introduction to Top players in Market – Google, MS, Amazon &Market trends Messaging Platforms (Facebook, WhatsApp) and Smart speakers – Alexa, Google Home and other new channels Ethical and Legal Considerations in AI Overview

Foundational Blocks for Programming Basic Python programming concepts, Node Basics Coding Best Practices Evaluation

Natural Language Processing:
Introduction: Brief history, Basic Concepts, Phases of NLP, Application of chatbotsetc General chatbot architecture, Basic concepts in chatbots: Intents, Entities, Utterances, Variables and Slots, Fulfillment Lexical Knowledge Networks (WordNet,Verbnet, PropBank, etc) Lexical Analysis, Part-of-Speech Tagging, Parsing/Syntactic analysis, Semantic Analysis, Word Sense Disambiguation. Information Extraction, Sentiment Analysis NLP using Python - Make use of any of the NLP libraries like NLTK, spaCy, Stanford NLP etc. (Practice session to use an NLP Tool -Hands on) Affective NLG.

Building a chatbot/Conversational AI Systems: Fundamentals of Conversational Systems (NLU, DM and NLG) Chatbot framework & Architecture, Conversational Flow & Design, Intent Classification (ML and DL based techniques), Dialogue Management Strategies, Natural Language Generation UX design, APIs and SDKs, Usage of Conversational Design Tools Introduction to popular chatbot frameworks – Google Dialog flow, Microsoft Bot Framework, Amazon Lex, RASA Channels: Facebook Messenger, Google Home, Alexa, WhatsApp, Custom Apps Overview of CE Testing techniques, A/B Testing, Introduction to Testing Frameworks -Botium /Mocha, Chai Security &Compliance – Data Management, Storage, GDPR, PCI, Building a Voice/Chat Bot - Hands on

Role of ML/AI in Conversational Technologies –Brief Understanding on how Conversational Systems uses ML technologies in ASR, NLP, Advanced Dialog management, Language Translation, Emotion/Sentiment Analysis, Information extraction ,etc. to effectively converse

Contact Centers: Introduction to Contact centers – Impact & Terminologies Case studies & Trends, How does a Virtual Agent/Assistant fit in here?

Overview on Conversational Analytics: Conversation Analytics: The need of it Introduction to Conversational Metrics

Future – Where are we headed? Summary, Robots and Sensory Applications overview**,** XR Technologies in Conversational Systems, XR-Commerce**,** What to expect next? – Future technologies and market innovations overview

Laboratory Work:
Project 1: Case Study to build a learning chatbot will ML based API like Amazon
Case Study to build a ML Model using LSTM/any RNN and integrate with chatbot

Course Learning Outcomes (CLOs) / Course Objectives (COs):
Upon completion of this course, students will be able to:

1. ML Concepts and different algorithms to build custom chatbot and conversational systems.
2. Comprehend the basics understanding of conversation environment.
3. Illustrate and apply the important features to describe a time series and perform simple analysis and computations on series.

Text Books:

1. Mactear, M. Conversational AI: Dialogue Systems, Conversational Agents, and Chatbots (Synthesis Lectures on Human Language Technologies (2020).
2. Edward Loper, Ewan Klein, and Steven Bird Natural Language Processing with Python Book (2002), 8th ed.
3. The Elements of Statistical Learning: Data Mining, Inference, and Prediction Trevor Hastie, Robert Tibshirani, and Jerome Friedman (2003) 6th ed.
4. The Hundred-Page Machine Learning Book Andriy Burkov.

Reference Books:

1. Machine Learning Tom M. Mitchell, Wiley-Interscience (2006), 4th ed.
2. Natural Language Processing in Action: Understanding, Analyzing, and Generating Text with Python Book by Cole Howard, Hannes Hapke, and Hobson Lane.